

RESEARCH ARTICLE

Inference on tree-structured subgroups with subgroup size and subgroup effect relationship in clinical trials

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Abstract

When multiple candidate subgroups are considered in clinical trials, we often need to make statistical inference on the subgroups simultaneously. Classical multiple testing procedures might not lead to an interpretable and efficient inference on the subgroups as they often fail to take subgroup size and subgroup effect relationship into account. In this paper, built on the selective traversed accumulation rules (STAR), we propose a data-adaptive and interactive multiple testing procedure for subgroups which can take subgroup size and subgroup effect relationship into account under prespecified tree structure. The proposed method is easy-to-implement and can lead to a more interpretable and efficient inference on prespecified tree-structured subgroups. Possible accommodations to post hoc identified tree-structure subgroups are also discussed in the paper. We demonstrate the merit of our proposed method by re-analyzing the panitumumab trial with the proposed method.

KEYWORDS

efficiency, interpretability, multiple testing, subgroup analysis

1 | INTRODUCTION

In clinical trials, subgroup analysis refers to the discovery and confirmation of treatment effect heterogeneity within a population. In modern drug development, subgroup analysis plays an increasingly important role as novel therapies are often found to benefit some patients more than others.¹ When this happens, subgroup analysis can help identify patients with beneficial treatment effects and facilitate follow-up drug developments and regulatory processes. For example, by subgroup analysis, panitumumab was confirmed effective and subsequently granted marketing authorization by the European Medicines Agency (EMA) in treating metastatic colorectal cancer (mCRC) patients with wild-type KRAS gene, while the initial filing of panitumumab for mCRC patients was declined by the EMA due to the low overall efficacy of panitumumab.² Recently, the U.S. Food and Drug Administration (FDA) approved atezolizumab as an effective first-line treatment for metastatic non-small cell lung cancer (NSCLC) in a subgroup where patients have high PD-L1 expression without mutations in the EGFR gene or ALK gene.³ As such cases keep abounding in clinical trials, how to make interpretable and efficient statistical inference on subgroups becomes an important question.

When multiple candidate subgroups are considered simultaneously in clinical trials, inference on subgroups might be viewed as a multiple testing problem where appropriate adjustments to p -values are often needed to avoid the inflation of false positives.⁴ Over the past decades, numerous methods have been proposed for multiple testing problems based on different criteria of false positives. For example, to control the family-wise error rate (FWER), single-step methods^{5,6} and sequential methods^{7,8} have been proposed, and to control the false discovery rate (FDR), the Benjamini–Hochberg

(BH) procedure⁹ has been proposed and has been refined by Benjamini and Yekutieli under dependence later.¹⁰ Other procedures for multiple testing, such as Storey's procedure,¹¹ the conditional local FDR¹² and independent hypothesis weighting (IHW),¹³ have also been considered. Despite the broad availability of existing multiple testing procedures, directly applying them to make inference on subgroups might suffer from a loss of interpretability and efficiency due to several issues inherited from the natures of subgroup analysis; in particular, (1) subgroup size and (2) subgroup effect relationship.

In this paper, subgroup size is referred to as the prevalence of a candidate subgroup in the overall population. Subgroup size is one of the main considerations in regulatory decision-making and post-marketing drug development. First, a new treatment is unlikely to get regulatory approval for a subgroup with small subgroup size even if the treatment is confirmed to be effective there, because it is difficult to draw a meaningful, interpretable and reliable conclusion about the efficacy and safety findings in a subgroup with low prevalence.¹⁴ Second, even if a treatment is granted approval in a subgroup with small subgroup size, the post-marketing benefits of the treatment will be limited because only a small proportion of patients can benefit from the treatment.¹⁵ Although subgroup size plays an important role in drug development and regulatory decision-making, most existing methods in subgroup analysis do not take subgroup size into account. The problems become more acute as the more patients a subgroup includes, the lower average treatment effect a subgroup tends to have.¹⁶ Therefore, without taking subgroup size into account, existing methods that pursue the enhanced treatment effect often favor candidate subgroups with small subgroup size. Recently, several methods^{15,17} have been proposed to account for subgroup size in the selection of promising subgroups, but they are not designed for inference on subgroups. Besides subgroup size, subgroup effect relationship, the intrinsic logical relationship between treatment effects of subgroups, is another issue that existing methods in subgroup analysis often fail to account for but bears practical importance for interpretability and efficiency. For example, when the overall population is a mixture of several subgroups, the effect of the overall population should be between the effects of the subgroups.¹⁸ Ignoring such relationships may lead to conservative inference and statistical efficiency loss on subgroups. What's worse, the statistical inference result may be contradictory and counter-intuitive without accounting for subgroup effect relationship; see Simpson's paradox for example.¹⁹ To take subgroup effect relationship into account, we need to impose appropriate constraints on the result from statistical inference, which helps avoid paradoxical results and increases the interpretability of inference results. While constrained multiple testing procedures have been considered,²⁰⁻²² they are tailored to particular relationships, such as the effect hierarchy principle and the effect heredity principle,²³ which are different from the subgroup effect relationship arising in clinical trials.

To bridge the gap, we propose a data-adaptive and interactive multiple testing procedure for prespecified tree-structured subgroups in clinical trials. The tree structure is one of the most commonly used frameworks in subgroup analysis due to its efficient segregation of population and clear interpretation.²⁴ In this paper, we focus on prespecified tree-structured subgroups, a widely adopted situation in subgroup analysis,²⁵ while possible accommodations to post hoc identified tree-structured subgroups are also discussed. Under tree structure, though numerous methods have been proposed to identify promising subgroups,²⁶⁻³⁰ methods to infer tree-structured subgroups, in particular, those appropriately accounting for subgroup size and subgroup effect relationship, are still lacking. Built on the selective traversed accumulation rules (STAR) proposed by Lei et al (2021),³¹ we propose STAR_{subgroup} to incorporate the information of subgroup size and subgroup effect relationship under prespecified tree structure. The proposed method is easy-to-implement and can lead to a more interpretable and efficient inference on prespecified tree-structured subgroups in clinical trials.

The remainder of this article is organized as follows. In Section 2, we revisit the STAR framework and propose a multiple testing procedure STAR_{subgroup} for prespecified tree-structured subgroups while taking subgroup size and subgroup effect relationship into account. In Section 3, we demonstrate the merit of the proposed method through simulations. In Section 4, we apply the proposed method to reanalyze the clinical trial data from panitumumab study and show how the proposed method can help make more appropriate and interpretable decisions in drug development. In Section 5, we summarize our work and discuss potential directions for future studies.

2 | METHODOLOGY

In this section, we consider a classical problem setting in subgroup analysis where candidate subgroups are arranged in a prespecified tree structure. Built on the STAR framework, we take subgroup size and subgroup effect relationship into account simultaneously and propose an interpretable and efficient multiple testing procedure for prespecified tree-structured subgroups.

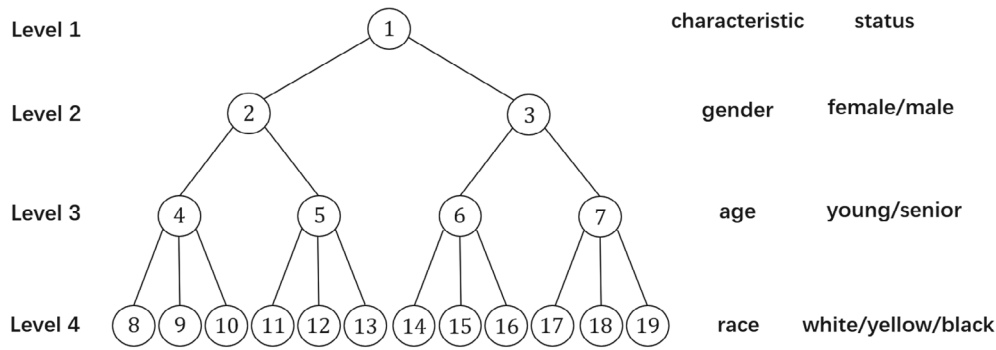


FIGURE 1 An illustrative example of prespecified tree-structured subgroups with four levels (i.e., $L = 4$). The root represents the overall population and each node (except the root) represents a subgroup of its parent node defined by the status of characteristics. For instance, the third node represents the male subgroup and the fifth node represents the senior female subgroup.

2.1 | Problem setting

Suppose there are k candidate subgroups arranged in a prespecified tree structure with L levels where a node represents a population and the topmost node represents the overall population. Except the nodes on the L -th level, each node is split into several child nodes representing disjoint subgroups of their parent node. The tree structure here can be prespecified based on clinical considerations or expert opinions. For illustration, Figure 1 provides a specific example of prespecified tree-structured subgroups with $L = 4$. Let μ_i and n_i denote the treatment effect and sample size of the i -th subgroup respectively for $i = 1, \dots, k$, where μ_i can be any reasonable measure of subgroup effects such as the difference of mean responses, as long as it satisfies some mild constraints in Section 2.3. Without loss of generality, we assume a larger μ_i indicates a better treatment effect, and our goal is to test whether patients within the subgroups derive no benefit from the treatment while taking subgroup size and subgroup effect relationship into account.

It is clear that the problem of interest can be viewed as a multiple testing problem

$$H_{0,i} : \mu_i \leq 0 \Leftrightarrow H_{1,i} : \mu_i > 0 \quad i = 1, \dots, k. \quad (1)$$

To control false positives in the above multiple testing problem, we consider the false discovery rate $\text{FDR} = \mathbb{E}[\text{FDP}]$ and aim for a prespecified significance level denoted by α , where $\text{FDP} = V / \max\{|\mathcal{R}|, 1\}$ denotes the false discovery proportion, $\mathcal{R} = \{i : H_{0,i} \text{ is rejected}\}$ denotes the rejection set, $\mathcal{H}_0 = \{i : H_{0,i} \text{ is true}\}$ denotes the true null hypotheses set, $V = |\mathcal{R} \cap \mathcal{H}_0|$ is the number of false rejections and $|\mathcal{R}|$ is the number of rejections. Let p_i denote the p -value for the i -th subgroup which can be calculated by any reasonable method based on the clinical trial data as long as it satisfies some mild assumptions as specified in Section 2.4. In this paper, we aim to figure out subgroups with significant treatment effects when accounting for subgroup size represented by relative sample size within the acceptable set $\mathcal{K}$, where $\mathcal{K}$ is a collection of subsets of $\{1, \dots, k\}$ satisfying two constraints with respect to subgroup effect relationship. The details of subgroup effect relationship and subgroup size will be specified in Section 2.3.

In this paper, we focus on inference on prespecified tree-structured subgroups, but the proposed method might accommodate the case of subgroups from tree-based subgroup identification methods. The reason is that to guarantee the validity of the proposed method, we mainly need to require p -values to satisfy the monotonicity assumption in Section 2.4. When the subgroups are identified from a separate dataset, a commonly used strategy in the practice of subgroup analysis,²⁷ they can be naturally regarded as prespecified subgroups and the monotonicity assumption is naturally satisfied. When the subgroups are identified from the same dataset as that used for inference, we acknowledge that the p -values might be overly optimistic and appropriate adjustment for the multiplicity in the tree structure construction might be needed. Fortunately, there exist several modern selective inference frameworks to adjust p -values appropriately to satisfy the p -value assumption in Section 2.4 for certain tree construction methods, such as tree values for classification and regression tree.³² With such adjusted p -values, we can implement our proposed method and deliver valid inference. The specific construction of the adjusted p -values is beyond the scope of this paper.

Algorithm 1. Selectively traversed accumulation rules (STAR)**Input:** p -values $\{p_i\}_{i=1}^k$, acceptable set $\mathcal{K}$ and target FDR level α **Output:** Final rejection set $\mathcal{R}$ Initialization: rejection set $\mathcal{R}_0 = \{1, \dots, k\}$ **while** $\mathcal{R}_t \neq \emptyset$ **do**

$$\widehat{\text{FDP}}_t = \frac{h(1) + \sum_{i \in \mathcal{R}_t} h(p_i)}{1 + |\mathcal{R}_t|}$$

if $\widehat{\text{FDP}}_t \leq \alpha$ **then return** $\mathcal{R}_t$ as the final rejection set $\mathcal{R}$ **else**Select a new rejection set $\mathcal{R}_{t+1} \subset \mathcal{R}_t$ with $\mathcal{R}_{t+1} \in \mathcal{K}$ based on $\{g(p_i)\}_{i=1}^k$ **end if****end while****2.2 | A revisit to selective traversed accumulation rules**

To start with, we revisit the selectively traversed accumulation rules (STAR) proposed by Lei et al (2021),³¹ which serve as a cornerstone of our proposed method. STAR is an interactive and flexible multiple testing procedure that can impose constraints on the rejection set under FDR control.

Selectively traversed accumulation rules (STAR) begin with a set consisting of all null hypotheses as an initial rejection set and iteratively shrink the rejection set subject to the constraints until the estimated FDP is below the target FDR level as illustrated in Algorithm 1. There are two main components in STAR: the accumulation function h and the masking function g . The accumulation function $h : [0, 1] \rightarrow [0, \infty)$ is a non-decreasing function with $\int_0^1 h(p) dp = 1$ and is used to estimate the FDP, and the masking function g is a function used to provide guidance on how to interactively shrink the rejection set subject to the constraints. How to choose $\mathcal{R}_{t+1}$ based on $\{g(p_i)\}_{i=1}^k$ depends on the specific construction of $g(p_i)$ and the purpose of the study. For example, if a larger $g(p_i)$ indicates a higher priority not to reject $H_{0,i}$, given $\mathcal{R}_t$, we might choose $\mathcal{R}_{t+1}$ as a subset of $\mathcal{R}_t$ which excludes a candidate hypothesis with the largest $g(p_i)$ in $\mathcal{R}_t$ without violating the constraints. More details about updating rejection sets in STAR can be found in sect. 3 of Lei et al (2021).³¹ Given an accumulation function h , to prevent FDR inflation, the masking function g needs to satisfy the mean-independent condition that $\mathbb{E}[h(p)|g(p)] \stackrel{a.s.}{=} \mathbb{E}[h(p)]$ for a null uniform p (see eq. 2 in Lei et al). For example, if $h(p) = 2 \cdot \mathbf{1}\{p \geq 0.5\}$, we can choose $g(p) = \min\{p, 1 - p\}$.

While STAR can enforce constraints in multiple testing, there exist several challenges to directly apply STAR to infer prespecified tree-structured subgroups. First, constraints with respect to subgroup effect relationship for tree-structured subgroups have not yet been specified. Second, STAR performs poorly when some null p -values concentrate near 1, which often arises in the one-sided test considered in subgroup analysis in Equation (1). Third, it is unclear how to take subgroup size into account when iteratively shrinking the rejection set. All the above call for a refinement of STAR to better serve the purpose of subgroup analysis.

2.3 | Proposed method

Built on STAR, we propose a multiple testing procedure $\text{STAR}_{\text{subgroup}}$ to infer prespecified tree-structured subgroups in clinical trials while taking subgroup effect relationship and subgroup size into account simultaneously. To address the limitations of STAR, we (1) define two subgroup effect constraints to boost the statistical power and ensure that the findings of the multiple testing follow the subgroup effect relationship and are interpretable; (2) generalize the masking function in STAR to address the one-sided test issue arising in subgroup analysis; and (3) establish a specific updating rule for the rejection set which can account for subgroup size and help deliver more interpretable findings. With the above three elements which will be detailed point by point later, $\text{STAR}_{\text{subgroup}}$ is summarized in Algorithm 2.

In $\text{STAR}_{\text{subgroup}}$, to take subgroup effect relationship into account, we consider the following two constraints for $\mathcal{K}$. Constraint 1 At least a child hypothesis of a rejected hypothesis must be rejected. Constraint 2 At least a child hypothesis of a non-rejected hypothesis must be non-rejected.

Algorithm 2. STAR_{subgroup}**Input:** p -values $\{p_i\}_{i=1}^k$, sample size $\{n_i\}_{i=1}^k$, acceptable set $\mathcal{K}$, target FDR level α and user-specific parameter r, θ **Output:** Final rejection set $\mathcal{R}$ Initialization: rejection set $\mathcal{R}_0 = \{1, \dots, k\}$ **while** $\mathcal{R}_t \neq \emptyset$ **do**

$$\widehat{\text{FDP}}_t = \frac{h(1) + \sum_{i \in \mathcal{R}_t} h(p_i)}{1 + |\mathcal{R}_t|}$$

if $\widehat{\text{FDP}}_t \leq \alpha$ **then return** $\mathcal{R}_t$ as the final rejection set $\mathcal{R}$ **else**Removable set of subgroups $\mathcal{C} = \{i \in \mathcal{R}_t : \mathcal{R}_t \setminus i \in \mathcal{K}\}$ **for** $i \in \mathcal{C}$ **do**

$$\text{Compute the subgroup score for the } i\text{-th subgroup } \text{Score}_i = r \frac{n_i}{\max_{i \in \mathcal{C}} \{n_i\}} - \frac{g^*(p_i)}{\max_{i \in \mathcal{C}} \{g^*(p_i)\}}$$

end forFind the subgroup $i_0 \in \mathcal{C}$ with the minimum subgroup scoreObtain a new rejection set $\mathcal{R}_{t+1} = \mathcal{R}_t \setminus i_0 \in \mathcal{K}$ **end if****end while**

In subgroup analysis, the above two constraints can be interpreted as that there is at least one effective subgroup in an effective population and there is at least one ineffective subgroup in an ineffective population respectively. Take Figure 1 as an example, if a therapy is effective in the overall population, it should be at least effective in either the male subgroup or the female subgroup. These constraints are motivated by the subgroup effect relationship that a population can be viewed as a mixture of its subgroups under tree structure and are naturally satisfied by many practical measures of subgroup effects in clinical trials, such as the difference of mean responses and log-relative response.^{18,33} However, due to several practical issues in study designs, such as the smaller sample sizes of subgroups and Simpson's paradox resulting from incomplete randomization in finite samples in clinical trials,³⁴ counter-intuitive inference result might arise in practice which is undesirable and motivates us to impose subgroup effect constraints on the rejection set. Although there exist several methods which might be used to partially address the count-intuitive results such as inverse probability weighting³⁵ and cluster analysis,³⁶ these methods are often designed for Simpson's paradox and are not directly applicable to the multiple testing problem considered in this paper. With the above two constraints, we wish to boost statistical power and deliver more interpretable results when inferring prespecified tree-structured subgroups with STAR_{subgroup}.

To address the issue in the one-sided test where some null p -values might be close to 1, we generalize the masking function in STAR based on the idea of asymmetries. In STAR, given an accumulation function h , the masking function is constructed as $g(p) = \min\{p, s(p)\}$ where $s(p)$ is the decreasing solution of $H(s(p)) = H(p)$ and $H(p) = \int_0^p \{h(x) - 1\} dx$. Such construction will lead to a symmetric masking function as illustrated in Figure 2, and a symmetric masking function might fail to distinguish very large p -values from the small ones and tend to overestimate FDP in the one-sided test considered in subgroup analysis. Recently, the AdaPT_g procedure³⁷ proposes an asymmetric masking function to reduce the bias in the FDP estimator when some null p -values concentrate near 1. Although the construction in AdaPT_g can not be applied to the STAR framework directly as it fails to meet the mean-independent condition which is crucial for FDR control under constraints, motivated by the idea of asymmetries in AdaPT_g, we propose the following generalized masking function $g^*(p)$ for STAR_{subgroup}

$$g^*(p) = \begin{cases} p & 0 \leq p \leq p^* \\ s(p) - \theta & p^* < p \leq s^{-1}(\theta) \\ s(p) + p^* - \theta & s^{-1}(\theta) < p \leq 1, \end{cases} \quad (2)$$

where p^* is the unique solution of equation $s(p) = p$, $\theta \in [0, p^*)$ controls the degree of asymmetries and $s^{-1}(\cdot)$ is the inverse function of $s(p)$. The existence of p^* and $s^{-1}(\cdot)$ is guaranteed by the monotonicity of $s(p)$ and a smaller θ indicates a weaker degree of asymmetries of $g^*(p)$. When $\theta = 0$, $g^*(p)$ reduces to the original masking function $g(p)$ considered in STAR. Similar to AdaPT_g, θ might be chosen by AIC or BIC in practice.³⁷ Under the asymmetric construction in Equation (2), the generalized masking function $g^*(p)$ can avoid treating small p -value and large p -value the same and can help eliminate

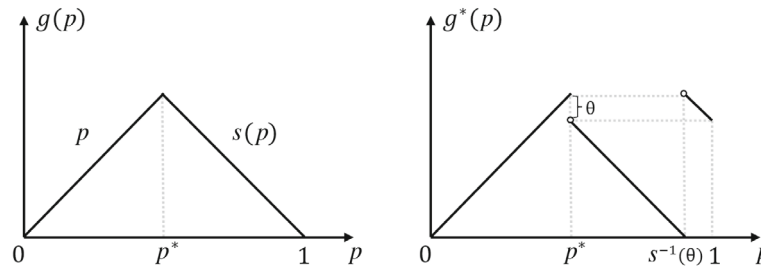


FIGURE 2 Illustration of the original masking function in STAR (left) and the generalized masking function in STAR_{subgroup} (right): The accumulation function is chosen as $h(p) = 2 \cdot \mathbf{1}\{p \geq 0.5\}$ so $s(p) = 1 - p$ and $p^* = 0.5$ with $\theta = 0.1$.

the overestimation of FDP in the presence of large p -values while satisfying the mean-independent condition. Figure 2 provides an illustration of how the asymmetries of the generalized masking function might help prevent large p -values from distorting the FDP estimation in the one-sided test with a specific example, and additional simulation results in Appendix B comparing our proposed method with the original STAR demonstrate the benefits of the generalized masking function.

In order to take subgroup size into account when iteratively updating the rejection set, we propose a subgroup score for the i -th subgroup with a non-negative user-specific parameter r and the removable set C as follows,

$$\text{Score}_i = r \frac{n_i}{\max_{i \in C} \{n_i\}} - \frac{g^*(p_i)}{\max_{i \in C} \{g^*(p_i)\}}. \quad (3)$$

Score_i is a linear combination of the masked p -value $g^*(p_i)$ and the sample size n_i after standardization, and a relatively larger Score_i indicates a higher priority to reject $H_{0,i}$, the null hypothesis that the treatment derives no benefit in the i -th subgroup. Because masked p -values indicate the strength of subgroup effects and relative sample size indicates subgroup size in clinical trials, the proposed subgroup score strikes a balance between subgroup effect and subgroup size via r which measures the relative importance associated with subgroup size. A larger r imposes more preference to the subgroups with high prevalence, and in practice, the choice of r might depend on how much importance the user would like to attach to subgroup size. For example, if subgroup size is regarded half as important as subgroup effect based on prior information such as regulatory requirements and post-marketing expectations, then we suggest choosing $r = 0.5$ as r is interpreted as the relative importance associated with subgroup size. In addition, we suggest choosing r between 0 and 1 to avoid excessive influence of subgroup size based on empirical evidence in Section 3. Although we adopt the additive form in Equation (3), the subgroup score might be transformed into other forms as long as the trade-off between subgroup effect and subgroup size is appropriately accounted for, such as a multiplicative form by exponential transformation.

In sum, as shown in Algorithm 2, STAR_{subgroup} proceeds by iteratively shrinking the rejection set based on the subgroup score that is built on the generalized masking function and accounts for subgroup size. The updating of the rejection set is restricted to the acceptable set $\mathcal{K}$ that reflects subgroup effect relationship and terminates when the estimated FDP falls below the target level.

2.4 | Justification

To justify the validity of the proposed method, we make the following three mild assumptions about the null p -values and the accumulation function:

Assumption 1 (Monotonicity). Each null p -value has a non-decreasing density.

Assumption 2 (Independence). The null p -values $\{p_i\}_{i \in H_0}$ are mutually independent and independent of the non-null p -values $\{p_i\}_{i \notin H_0}$, conditional on the sample size $\{n_i\}_{i=1}^k$.

Assumption 3 (Desirability). The accumulation function $h(p)$ is a constant when $p < p_0$, where $p_0 = \sup\{p : h(p) \leq 1\}$.

Assumption 1 is slightly stronger than the usual assumption that the null p -value is stochastically larger than uniform distribution and is satisfied by most practical testing procedures for prespecified subgroups in subgroup analysis. When subgroups are identified from tree-based subgroup identification methods, the apparent p -values might be overly optimistic and might not satisfy Assumption 1. In this case, we might adopt modern selective inference frameworks to adjust the p -values appropriately and implement the proposed methods with the adjusted p -value instead as discussed in Section 2.1. Assumption 2 requires that the p -values of different subgroups are independent, which can be naturally satisfied when patients on different levels are collected from different clinical trials, such as the pooled analyses of multiple trials.³⁸ Although the independence assumption may not be satisfied when subgroup analysis is conducted on a single trial, we note that the p -values of subgroup effect size can only be either independent or positively correlated due to the inherent nature of subgroups. Similar to many existing multiple testing procedures, without theoretical guarantees, STAR_{subgroup} still performs well numerically when p -values are positively correlated as demonstrated in the simulation results in Section 3, though several commonly seen technical gaps often hinder us from establishing theoretical guarantees of multiple testing procedures in the presence of correlated p -values.³⁹ Assumption 3 ensures that the accumulation function is desirable to estimate FDP, which is satisfied by most accumulation functions used in the existing literature.^{40,41}

Theorem 1. *Under Assumptions 1, 2, and 3, when the generalized masking function $g^*(p)$ is constructed by Equation (2), the FDR of STAR_{subgroup} is not larger than α conditional on $\{n_i\}_{i=1}^k$.*

Theorem 1 states that under Assumptions 1, 2, and 3, the proposed algorithm STAR_{subgroup} can control FDR at the target level with the proposed generalized masking function in Equation (2). The proof of Theorem 1 and more details about the existence and construction of $s(p)$ are provided in Appendix A.

3 | SIMULATION STUDIES

In this section, we conduct Monte Carlo simulations to evaluate the performance of STAR_{subgroup} in finite samples. We focus on the scenario where subgroups are arranged in a prespecified tree with $L = 6$, a reasonable number of levels considered in the current practice of subgroup analysis.^{42,43} Except those on the bottom level of the tree, each population is split into two child subgroups with equal sample sizes. Let $n_1 = 1600$ be the number of patients in the overall population. Then, the sample size of the i -th subgroup is $n_i = \frac{1600}{2^{L_i-1}}$ for $i = 1, \dots, k$ where L_i is the level of the i -th subgroup and $L_1 = 1$. As illustrated in Figure 3, we consider three representative situations where treatment effect heterogeneity arises at different levels of the tree, and for each representative situation, we apply the proposed method STAR_{subgroup} to multiple trials case where the independence assumption is satisfied and single trial case where the independence assumption is violated in Sections 3.1 and 3.2 respectively.

Throughout the simulation studies, we consider the same accumulation function $h(p) = 2 \cdot \mathbf{1}\{p \geq 0.5\}$ as that in Lei et al (2021) and the corresponding generalized masking function proposed in Equation (2)

$$g^*(p) = \begin{cases} p & 0 \leq p \leq 0.5 \\ 0.9 - p & 0.5 < p \leq 0.9 \\ 1.4 - p & 0.9 < p \leq 1, \end{cases} \quad (4)$$

where θ is set to 0.1 based on empirical evidence. The results for other values of θ are discussed in Appendix B. For each situation in Figure 3, we generate 1000 Monte Carlo samples and evaluate the performance in terms of three quantities of interest:

- False discovery rate (FDR): $\mathbb{E}[|\mathcal{R} \cap \mathcal{H}_0| / \max\{|\mathcal{R}|, 1\}]$, which measures the ability of a method to control false positives in multiple testing;
- Statistical power: $\mathbb{E}[|\mathcal{R} \cap \mathcal{H}_1| / \max\{|\mathcal{H}_1|, 1\}]$ with $\mathcal{H}_1 = \mathcal{R} \setminus \mathcal{H}_0$, which measures the ability of a method to detect effective subgroups correctly in multiple testing;

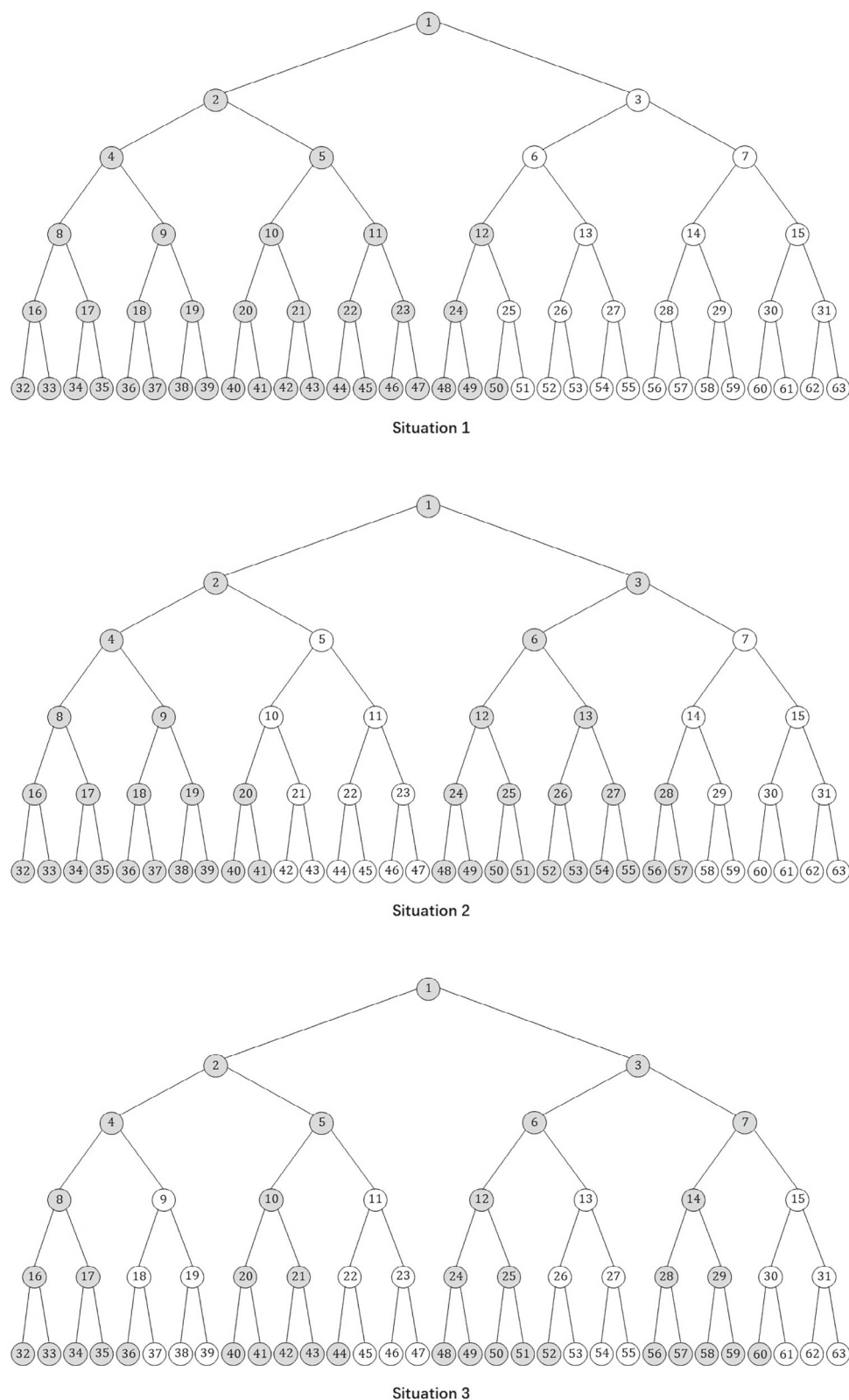


FIGURE 3 Three representative situations of prespecified tree-structured subgroups considered in the simulation where treatment effect heterogeneity arises at level 1, level 2, and level 3, respectively. Each node represents a population with grey ones corresponding to non-nulls and white ones corresponding to nulls.

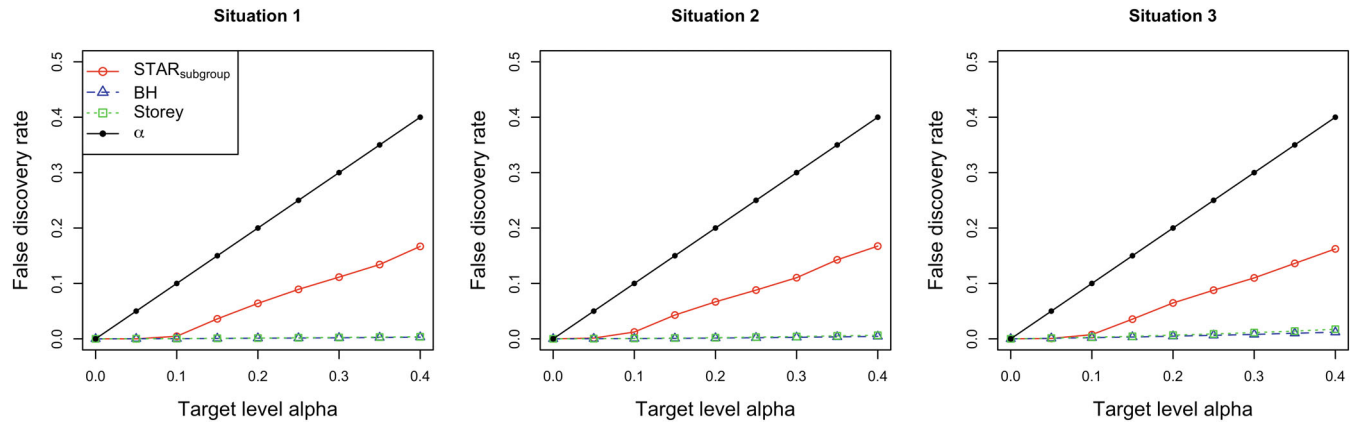


FIGURE 4 Comparison of FDR of $\text{STAR}_{\text{subgroup}}$ (red solid), the BH procedure (blue dashed) and Storey's procedure (green dotted) in different target levels with fixed $r = 0$ based on multiple trials, where the black solid line is the target level α .

- Power to detect the effective subgroup with the largest subgroup size (or power to detect the largest for short): $\mathbb{E}[|\mathcal{R} \cap \mathcal{H}_{\max}| / \max\{|\mathcal{H}_{\max}|, 1\}]$ with $\mathcal{H}_{\max} = \{i : i \in \mathcal{H}_1, n_i = \max_{i \in \mathcal{H}_1} n_i\}$, which measures the ability of a method to detect effective subgroups with high prevalence in multiple testing.

For the purpose of comparison, we include the Benjamini–Hochberg (BH) procedure⁹ and Storey's procedure¹¹ in simulations, which are commonly used multiple testing procedures for FDR control. In single trial case, we also include the Benjamini–Yekutieli (BY) procedure,¹⁰ a modification of the BH procedure for the dependent scenario, as an additional comparison, where the conclusion is similar to the BH procedure and is presented in Appendix B.

3.1 | Multiple trials

To start with, we consider inference on prespecified tree-structured subgroups based on multiple trials which often arise in pooled analyses.³⁸ When the p -values of subgroups on different levels are calculated from different trials, the independence assumption in Theorem 1 is naturally satisfied. To mimic the case of multiple trials, in the i -th subgroup, we randomly assign n_i patients into the treatment group and the control group with equal probability and generate independent observations from the normal distribution $N(\mu_i, 2.5^2)$ for the patients in the treatment group and from $N(0, 2.5^2)$ for those in the control group. On the bottom level, the non-null μ_i 's are set to be 1.5 and the null μ_i 's are set to be -2 to satisfy the three representative situations. On the upper levels, the μ_i 's are determined by the average of the treatment effects of their child subgroups weighted by the relative sample sizes. In particular, for the binary tree considered in our simulations, μ_i is equal to $\frac{1}{2}(\mu_{2i} + \mu_{2i+1})$ for $i = 1, \dots, 2^{L-1} - 1$. The p -value p_i is calculated by the one-sided t -test on the average treatment effect with the null hypothesis $H_{0,i} : \mu_i \leq 0$. The results are summarized in Figures 4 and 5.

From Figure 4, we see that $\text{STAR}_{\text{subgroup}}$ controls FDR across different target levels. For the sake of illustration, we fix $r = 0$ here, and the FDR of the proposed method with other values of r is similar and presented in Appendix B. While the BH procedure and Storey's procedure also control FDR well as illustrated in Figure 4, Figure 5 shows that fixing $\alpha = 0.15$, $\text{STAR}_{\text{subgroup}}$ is more powerful than the BH procedure and Storey's procedure across a broad range of r . The reason is that different from the BH and Storey's procedure, $\text{STAR}_{\text{subgroup}}$ can exploit the information of subgroup effect relationship. Figure 5 also suggests that as we increase r to impose more emphasis on subgroup size, $\text{STAR}_{\text{subgroup}}$ gains remarkable power to detect the largest without losing too much power in general.

3.2 | Single trial

Next, we consider inference on prespecified tree-structured subgroups based on a single trial. In subgroup analysis, the patient data for different subgroups might be collected from the same initial trial. When this happens, there exist positive correlations between the p -values of the parent subgroups and those of their child subgroups and the independence

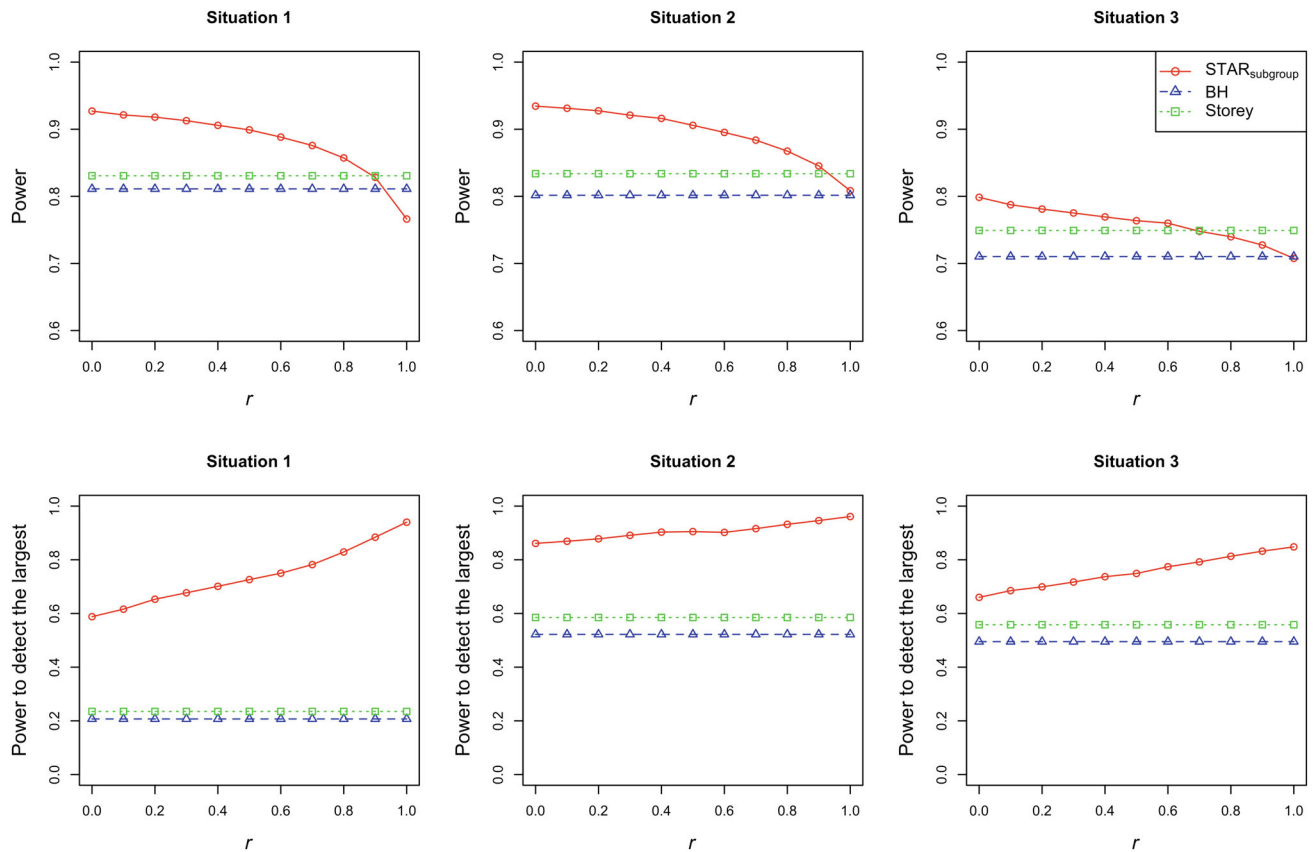


FIGURE 5 Impact of r : Power and power to detect the largest of $\text{STAR}_{\text{subgroup}}$ (red solid), the BH procedure (blue dashed) and Storey's procedure (green dotted) in different r 's with fixed target level $\alpha = 0.15$ based on multiple trials.

assumption in Theorem 1 is not necessarily satisfied. To examine the performance of the proposed method in this scenario, we keep the same data generation scheme for the subgroup on the bottom level as that in Section 3.1, but on the upper levels, we let the observations in a population consist of observations in its child subgroups. Similarly, the p -value p_i is calculated by the one-sided t -test on the average treatment effect with the null hypothesis $H_{0,i} : \mu_i \leq 0$.

Figure 6 shows the FDRs of $\text{STAR}_{\text{subgroup}}$, the BH procedure and Storey's procedure at different target levels, and Figure 7 shows the power and power to detect the largest of these two procedures in different r 's. From Figures 6 and 7, we see that $\text{STAR}_{\text{subgroup}}$ still controls the FDR at the target level though the theoretical guarantee is lacking, which supports the practical use of $\text{STAR}_{\text{subgroup}}$ in the case of single trial. Moreover, $\text{STAR}_{\text{subgroup}}$ maintains higher power and power to detect the largest across different r 's.

We also conduct more simulations to demonstrate that (1) the proposed method controls FDR across a broad range of r ; (2) the generalized masking function helps improve statistical power and power to detect the effective subgroup with the largest subgroup size; (3) the proposed method also outperforms the modification of the BH procedure for positively correlated p -values; (4) the proposed method maintains good performance even when the tree is of few levels and low complexity; (5) the proposed method performs well when θ is appropriately chosen and (6) the enhanced power of the proposed method to detect effective subgroups with large subgroup size is not at the cost of increasing the type-I-error rate in large subgroups. Please refer to Appendix B for details about the additional simulation studies.

4 | APPLICATION TO PANITUMUMAB TRIAL DATA

In this section, we revisit the clinical trial data of the panitumumab study to demonstrate the benefits of the proposed method. Panitumumab, a fully human monoclonal antibody directed against the epidermal growth factor receptor

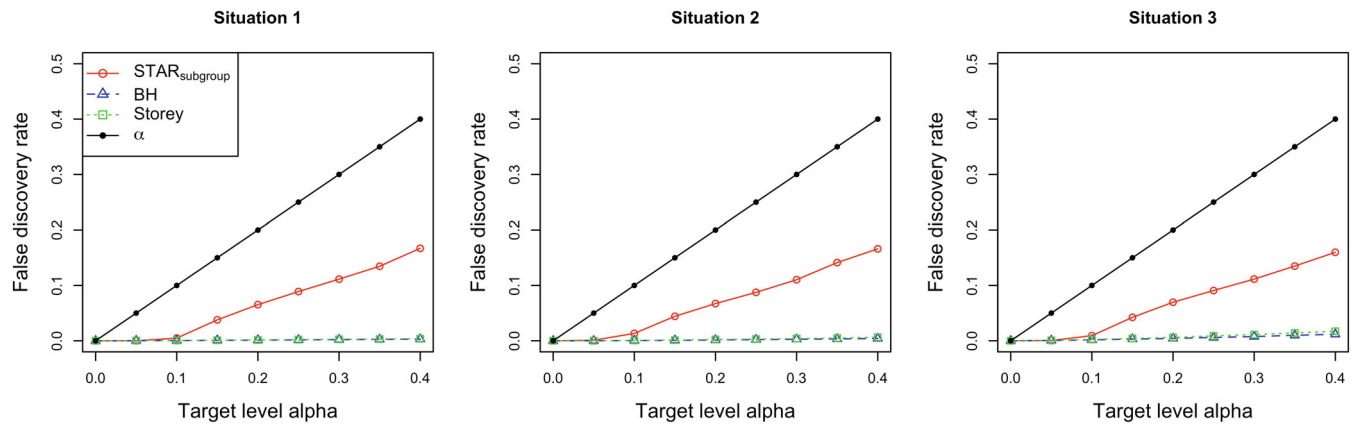


FIGURE 6 Comparison of FDR of STAR_{subgroup} (red solid), the BH procedure (blue dashed) and Storey's procedure (green dotted) in different target levels with fixed $r = 0$ based on a single trial, where the black solid line is the target level α .

(EGFR), is a treatment for patients with metastatic colorectal cancer (mCRC). In the initial phase III trial of panitumumab reported by Van Cutsem et al, although progression-free survival (PFS) was significantly prolonged, no difference in overall survival was observed and the difference in median PFS was relatively small for panitumumab in the overall population of the patients with mCRC.⁴⁴ Thus, the initial filing of panitumumab for mCRC patients was declined by the European Medicines Agency (EMA).

After the completion of the initial trial, a biomarker called KRAS was found to be correlated with the poor response to panitumumab in patients with mCRC.⁴⁵ KRAS is an oncogene that encodes a small G-protein acting on the EGFR. Mutant KRAS yields a constitutively active protein that might cause a lack of response to EGFR inhibitors, such as panitumumab, and it is thus guessed that only patients with wild-type KRAS might benefit from panitumumab. Motivated by that, Amado et al conducted a subgroup analysis of panitumumab and found that the treatment effect of panitumumab in the wild-type KRAS subgroup was significantly better than those of other subgroups, such as the mutant-type KRAS subgroup.⁴⁵ With a follow-up confirmatory trial, panitumumab was confirmed effective and subsequently granted conditional marketing authorization by the EMA in treating mCRC patients with wild-type KRAS.²

Here, to demonstrate the benefits of STAR_{subgroup} in clinical trials, we revisit the data reported in the confirmatory trial of the panitumumab study with STAR_{subgroup}, a phase III trial to confirm the efficacy of FOLFIRI plus panitumumab compared with FOLFIRI alone for second-line treatment of mCRC in the wild-type KRAS subgroup.⁴⁶ This trial was modified from a trial for the overall population of 851 patients which allows us to look into other subgroups besides wild-type KRAS. We focus on four baseline characteristics: KRAS status (mutant vs. wild), gender (female vs. male), age (<65 vs. ≥ 65) and sites of metastatic disease or metsite for short (<3 vs. ≥ 3), and build a binary tree by recursively partitioning the patient population into two subgroups based on each characteristic status as shown in Figure 8. In practice, the splitting order might be determined by prior knowledge²¹ or tree-based subgroup identification methods.⁴⁷ Here, to demonstrate how the proposed method might help confirm the efficacy of panitumumab in the wild-type KRAS subgroup, we adopt a prespecified tree structure where we first split the overall population into the mutant KRAS subgroup and the wild-type KRAS subgroup by KRAS status. For the remaining splits, we start from basic demographical variables to more complex clinical features and choose the order “gender $\rightarrow$ age $\rightarrow$ metsite”. For other splitting orders, the results are summarized in Appendix B and we have similar conclusions that the proposed method can lead to more interpretable and efficient inference results in the panitumumab trial. We consider PFS with censoring information as the end point and the censoring rate is about 5% in this study. To make it consistent with the multiple testing problem considered in Equation (1), we use the negative log-hazard ratio defined by Cox proportional hazard model as the treatment effect, a positive treatment effect indicates that panitumumab-FOLFIRI is more effective than FOLFIRI alone, and the p -values are calculated by the one-sided t -test. For STAR_{subgroup}, the accumulation function is chosen as $h(p) = 2 \cdot \mathbf{1}\{p \geq 0.5\}$ and the generalized masking function is constructed by Equation (2) with $\theta = 0.15$. For the purpose of comparison, we also analyze the data with the BH procedure, and we fix the target FDR level as 0.2, the smallest FDR level to guarantee that the BH procedure has a non-empty rejection set.

From Figure 8, the BH procedure finds that panitumumab is effective in the third subgroup (wild-type KRAS subgroup), whereas it is ineffective in both of its child subgroups, which is clearly hard to interpret. By comparison, the result

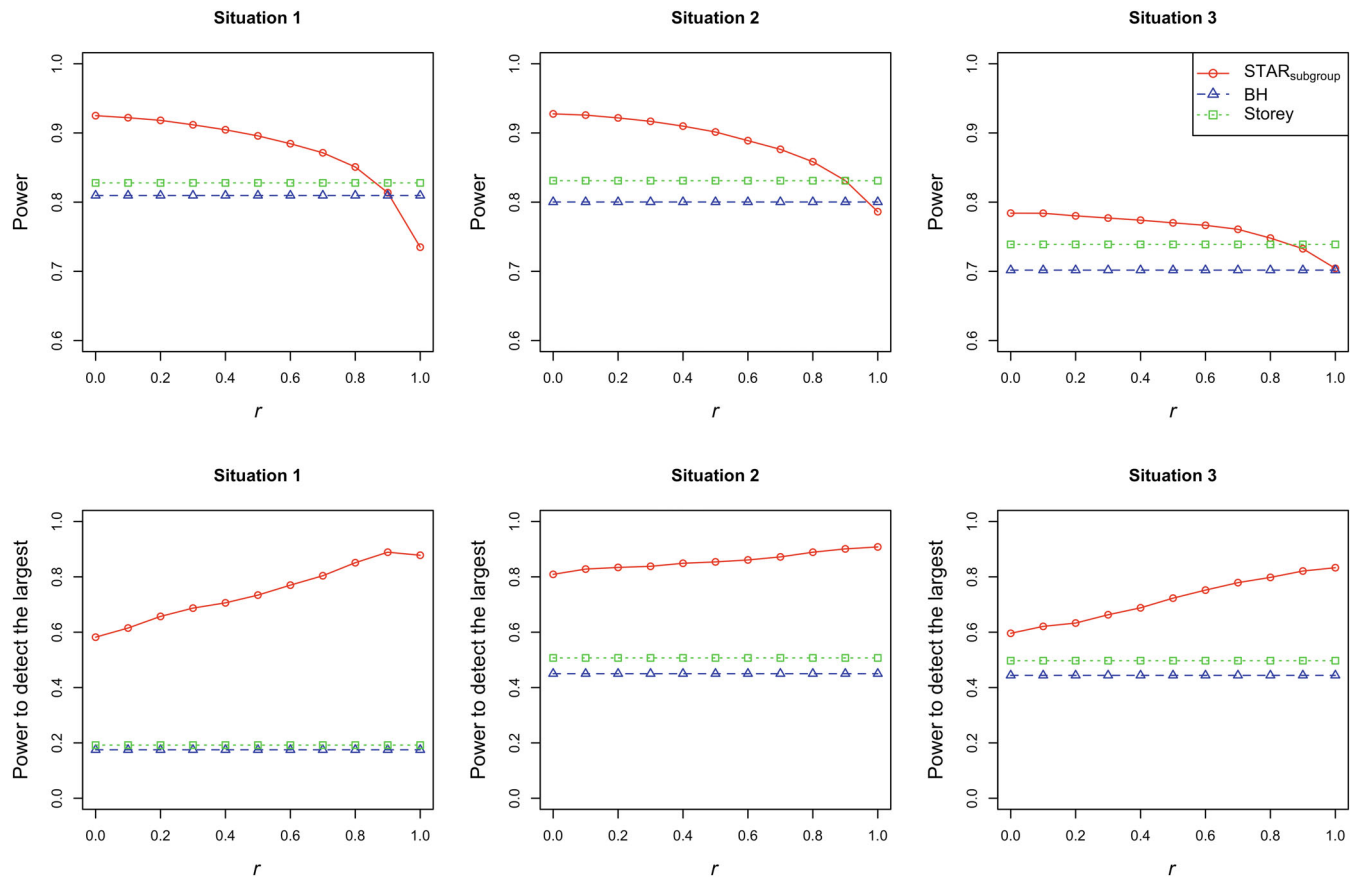


FIGURE 7 Impact of r : Power and power to detect the largest of $\text{STAR}_{\text{subgroup}}$ (red solid), the BH procedure (blue dashed) and Storey's procedure (green dotted) in different r 's with fixed target level $\alpha = 0.15$ based on a single trial.

from $\text{STAR}_{\text{subgroup}}$ is easier to interpret as it takes subgroup effect relationship into account and has two clear paths starting from the wild-type KRAS subgroup, which might indicate the drug mechanism of panitumumab and be worth further study. Both multiple testing procedures find that panitumumab is effective in the overall population, which might be due to the relatively large sample size of the overall population as this trial as well as its sample size is modified for the wild-type KRAS subgroup. To better illustrate the benefit of our proposed method, we consider a subsample of panitumumab trial data.

To have a subsample, we focus on the patients with greater disease severity which can be determined by the ECOG performance status. The ECOG performance status is widely used by physicians and researchers to assess the disease progression of patients and determine appropriate treatment, and a positive ECOG performance status indicates that the disease has affected the daily living abilities of patients, which is common in the clinical treatment of metastatic colorectal cancer. As shown in Figure 9, we analyze the performance of the proposed method on 300 patients with positive ECOG performance status. From Figure 9, we can see that only $\text{STAR}_{\text{subgroup}}$ finds that panitumumab is effective for mCRC patients with wild-type KRAS, which is more in line with the current belief from the existing studies of panitumumab and demonstrates the benefit of taking subgroup size into account in identifying the large effective subgroup.

5 | DISCUSSION

In this paper, we propose a data-adaptive and interactive multiple testing procedure for prespecified tree-structured subgroups in clinical trials based on the STAR framework. The proposed method can account for subgroup size and subgroup effect relationship simultaneously, which is not only applicable to prespecified tree-structured subgroups but

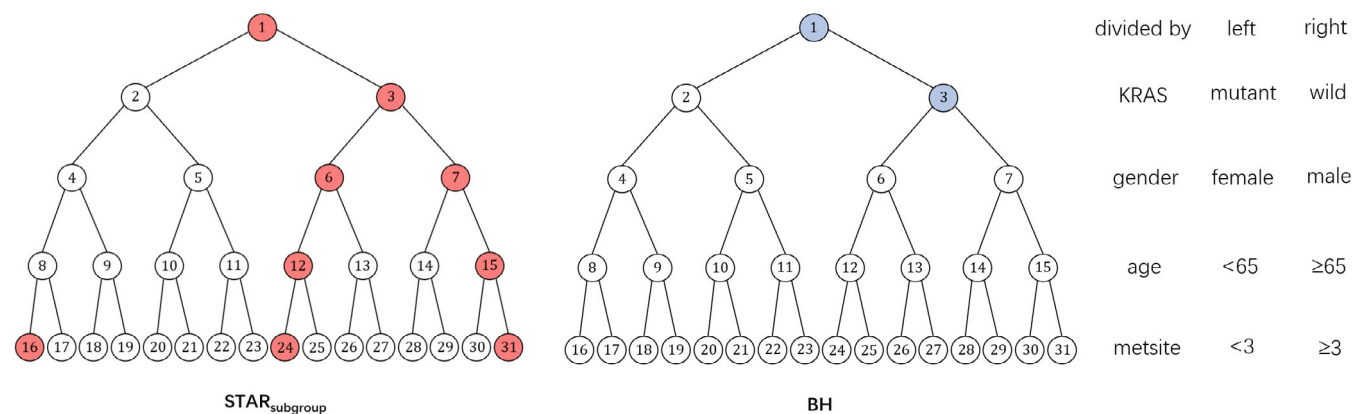


FIGURE 8 The rejection sets of STAR_{subgroup} with $r = 0.2$ (red) and the BH procedure (blue) with fixed target level $\alpha = 0.2$ when applied to panitumumab trial data.

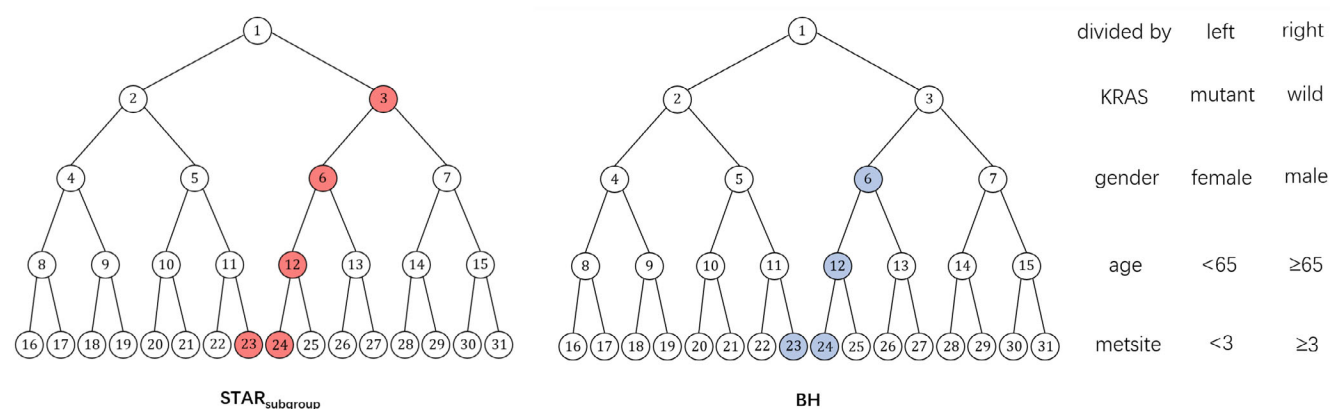


FIGURE 9 The rejection sets of STAR_{subgroup} with $r = 0.2$ (red) and the BH procedure (blue) with fixed target level $\alpha = 0.35$ when applied to the subsample with a positive ECOG performance status of panitumumab trial data.

might also accommodate post hoc identified tree-structured subgroups. The salient point of this work is that appropriate inference on subgroups needs to take subgroup size and subgroup effect relationship into account, which classical multiple testing procedures often ignore. The proposed method is easy-to-implement and leads to a more interpretable and efficient inference on prespecified tree-structured subgroups. We revisit the panitumumab trial with the proposed method and show that the findings from the proposed method are more consistent with the existing studies of panitumumab.

The proposed method focuses on tree-structured subgroups. Although tree structure is widely adopted in subgroup analysis, in practice, we might also encounter situations where candidate subgroups can not be arranged in a tree, such as female subgroup and senior subgroup. To deal with such general overlapping subgroups, a possible approach is to separate general overlapping subgroups into non-overlapping subgroups and connect their subgroup effects with linear transformation as described in Guo et al.⁴⁸ In this case, we need to specify subgroup effect constraints in a more general scope that can account for the linear transformation and develop appropriate multiple testing procedures for subgroups to take subgroup size and subgroup effect relationship into account beyond tree structure, which we would like to leave to future studies.

While the proposed method is developed to take subgroup size and subgroup effect relationship into account in subgroup inference, our framework might be extended to take other practical considerations, such as the cost of treatment, into account when evaluating the treatment effect. The extension of the proposed framework to address other practical questions beyond subgroup analysis shall be investigated in the future.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Project Data Sphere at <https://doi.org/10.34949/0ws2-w454>, reference number 46.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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